

Multiple integrals – 3

Volumes, sections and asymptotics in the dimension

Homework

13 сентября 2026 года

Theoretical minimum. A volume is the integral of the areas of the sections, and this single remark computes the volume of the ball, of the simplex and of most sets defined by inequalities between the coordinates. In high dimension the picture is counterintuitive: the volume of the unit ball tends to zero, the ratio to the cube tends to zero as well, and almost all of the volume sits in a slab of width $1/\sqrt{n}$ around any equator. Volumes of subsets of the cube are probabilities for independent uniform coordinates, so order statistics and volume computations are the same subject; the volume of a parallelepiped is the square root of the Gram determinant. Inequalities between volumes belong to the series on convex geometry; here the volumes are computed.

1. Compute the volume of the simplex $\{0 \leq x_1 \leq \dots \leq x_n \leq 1\}$ in two ways — by Cavalieri and by symmetry over the $n!$ orderings — and deduce the density of the k -th order statistic of n uniform points.
2. Find the volume of $\{x \in \mathbb{R}^n : |x_1| + \dots + |x_n| \leq 1\}$, of $\{x : \|x\|_p \leq 1\}$ for $p > 0$, and the asymptotics of the last one as $n \rightarrow \infty$ for fixed p .
3. Prove that the surface area of the unit sphere in $\mathbb{R}^n$ equals nV_n , find the dimension in which it is maximal, and prove that $\frac{\sigma(S^{n-1})}{V_n} \rightarrow \infty$.
4. Prove that the relative volume of the slab, that is $\frac{\text{vol}(B_n \cap \{|x_1| \leq c/\sqrt{n}\})}{\text{vol } B_n}$, tends to $\frac{1}{\sqrt{2\pi}} \int_{-c}^c e^{-s^2/2} ds$ as $n \rightarrow \infty$; along the way prove that $\frac{V_{n-1}}{V_n \sqrt{n}} \rightarrow \frac{1}{\sqrt{2\pi}}$.
5. Compute the volume of the set $\{x \in [0, 1]^n : \max_i x_i - \min_i x_i \leq t\}$ and find its limit behaviour as $n \rightarrow \infty$ for fixed t .

6. Prove that among all parallelepipeds spanned by vectors of given lengths the volume is maximal exactly for orthogonal vectors (Hadamard's inequality, Linear algebra – 5), and give the volume interpretation of the Gram determinant of linearly dependent vectors.
7. Compute the volume of the region bounded by the surface $(\frac{x}{a})^{2/3} + (\frac{y}{b})^{2/3} + (\frac{z}{c})^{2/3} = 1$ and the volume of the solid of revolution of the astroid.
8. Let $K \subset \mathbb{R}^n$ be convex, symmetric and of volume greater than 2^n . Prove that K contains a non-zero lattice point (Minkowski, proved in Convex – 10), and give the volume-counting proof: compare $\text{vol}(K/2)$ with the number of lattice translates.
9. Compute the volume of the convex hull of the unit ball in $\mathbb{R}^3$ and a point at distance $h > 1$ from its centre, and find the limit of the ratio of this volume to the volume of the cone with the same apex and base as $h \rightarrow \infty$.
10. Prove that for a bounded measurable $A \subseteq \mathbb{R}^n$ the function $t \mapsto \text{vol}(A \cap \{x_1 < t\})$ is continuous and non-decreasing, deduce the ham-sandwich statement in dimension one, and explain how Cavalieri gives the general bisection argument.